

Buprenorphine Physician-Pharmacist Collaboration in the Management of Patients With Opioid Use Disorder

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1. Background and Rationale

Opioid use disorder (OUD) is a major public health crisis in the United States, with more than 2 million people meeting diagnostic criteria and opioid overdose deaths continuing to rise. Buprenorphine, a partial mu-opioid agonist approved by the FDA for the treatment of OUD (Schedule III), significantly reduces illicit opioid use, overdose risk, and mortality when prescribed appropriately. However, access remains severely constrained by the limited number of DATA-waivered prescribers and structural barriers in primary care settings.

Collaborative Practice Agreements (CPAs) between physicians and clinical pharmacists have demonstrated effectiveness in improving outcomes for chronic diseases including hypertension, diabetes mellitus, and anticoagulation therapy. CTN-0075 tests whether a structured physician-pharmacist collaboration model—formalized via CPA and delivered within existing healthcare infrastructure—can improve retention in buprenorphine treatment for OUD.

2. Objectives

2.1 Primary Objective

To evaluate the effect of a physician-pharmacist collaborative care model versus treatment as usual on retention in buprenorphine treatment for OUD at 6 months post-randomization.

2.2 Secondary Objectives

- Assess illicit opioid use (urine drug screen and Timeline Followback) at 1, 3, and 6 months.
- Evaluate patient-reported quality of life (SF-12), treatment satisfaction, and opioid craving (VAS).
- Assess healthcare utilization (ED visits, hospitalizations) over the 6-month treatment period.
- Examine the impact of the CPA model on buprenorphine dose adjustments and medication possession ratio (MPR).
- Evaluate cost-effectiveness of the collaborative care model.

3. Study Design

CTN-0075 is a pragmatic, parallel-group, randomized controlled trial conducted at approximately 10 NIDA Clinical Trials Network (CTN) affiliate sites across the United States. Participants are randomized 1:1 to the physician-pharmacist collaborative model (intervention) or treatment as usual (control). Randomization is stratified by site and baseline OUD severity (moderate vs. severe).

Parameter	Detail
Design	Parallel-group randomized controlled trial
Allocation ratio	1:1 (intervention : control)
Blinding	Open-label; assessor-blinded primary outcome verification
Active treatment period	6 months
Follow-up period	3 months post-treatment
Target enrollment	1,000 participants (500 per arm)
Number of sites	~10 NIDA CTN-affiliated sites
Primary endpoint	Retention in buprenorphine treatment at Month 6

4. Eligibility Criteria

4.1 Inclusion Criteria

- Age ≥ 18 years at enrollment.
- DSM-5 diagnosis of moderate or severe opioid use disorder.
- Currently receiving or initiating buprenorphine/naloxone at an enrolled study site.
- Willing and able to provide written informed consent.
- Has a primary care or addiction medicine provider at the participating site.

4.2 Exclusion Criteria

- Current enrollment in a methadone maintenance treatment program.
- Active suicidal ideation requiring immediate psychiatric intervention.
- Severe hepatic impairment (Child-Pugh Class C) contraindicating buprenorphine use.
- Pregnancy at enrollment or planning to become pregnant during the study (separate pregnancy sub-study available).
- Cognitive impairment precluding ability to provide informed consent.
- Prior participation in CTN-0075 or concurrent enrollment in a conflicting clinical trial.

5. Interventions

5.1 Physician-Pharmacist Collaborative Model (Intervention)

Participants in the intervention arm receive buprenorphine managed through a formalized CPA between their prescribing physician and a credentialed clinical pharmacist. The pharmacist conducts monthly Medication Therapy Management (MTM) sessions (in-person or telephonic), monitors adherence via prescription fill records, adjusts buprenorphine doses within physician-approved protocols, orders and reviews urine drug screens (UDS), and communicates with the prescriber via structured EHR-embedded notes.

5.2 Treatment as Usual (Control)

Control arm participants receive standard buprenorphine prescribing as practiced at the enrolling site. Routine pharmacist dispensing interactions may occur but no formalized CPA-based collaborative management is provided.

6. Outcome Measures

6.1 Primary Outcome

Retention in buprenorphine treatment at Month 6, defined as filling at least one buprenorphine prescription in the 30-day window centered on the 6-month assessment.

6.2 Secondary Outcomes

Outcome	Instrument/Method	Timepoint(s)
Illicit opioid use	Urine drug screen (UDS)	Months 1, 3, 6
Self-reported opioid use	TLFB (Timeline Followback)	Months 1, 3, 6
Opioid craving	Visual Analog Scale (0–100 mm)	Months 1, 3, 6
Quality of life	SF-12 (PCS + MCS)	Baseline, Months 3, 6
Treatment satisfaction	5-item Likert scale	Months 3, 6
Healthcare utilization	Self-report + EHR extraction	Month 6
Buprenorphine adherence	Medication Possession Ratio	Month 6

7. Statistical Considerations

7.1 Sample Size

Assuming 55% retention in the control arm and 65% in the intervention arm (10% absolute improvement), with 80% power and two-sided $\alpha = 0.05$, 388 participants per arm are needed. Inflating for 10% attrition yields a target of 500 per arm (1,000 total).

7.2 Primary Analysis

Intent-to-treat (ITT) analysis. Retention at Month 6 compared using logistic regression adjusting for randomization strata. Missing outcomes handled by multiple imputation under a missing-at-random assumption, with sensitivity analyses assuming missing = not retained.

8. Safety Monitoring

An independent Data and Safety Monitoring Board (DSMB) reviews unblinded safety and efficacy data at pre-specified interim analyses. Serious adverse events (SAEs) must be reported to NIDA CTN leadership, the DSMB chair, and the local IRB within 7 calendar days. The trial may be stopped early for safety, efficacy, or futility per pre-specified rules defined in the DSMB charter.

9. Ethics and Regulatory

The protocol is approved by a central IRB operating under a reliance agreement with each participating site. All participants provide written informed consent prior to any study procedure. The study is conducted in accordance with ICH E6 Good Clinical Practice (GCP), applicable federal regulations (45 CFR Part 46, 21 CFR Parts 50 and 56), and NIDA CTN policies.